

Ryan Zheng

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Skills

CAD

- **SolidWorks** used on various projects involving **GD&T**, **FEA**, and **plastic injection molding**
- **AutoCAD** and **SolidWorks** used to draft 2D engineering drawings with **FAI** dimensions
- Experience in both low and high-volume manufacturing methods

Mechanical

- **3D Printing** and **Laser Cutting** used for rapid prototyping
- Created 3D models based on **DFMA** principles
- Used **Mastercam**, **MachPRO**, and **CMM** to fabricate and validate design
- Implemented **Root Cause Analysis**, **DFMEA** and **FACA** to address design issues

Software & Electrical

- C++, MATLAB, ANSYS
- Soldering, Wiring

Projects

Thermamount

SolidWorks, Ansys, CNC, Machine Tools

- Designed and deployed a weather resistant mounting and enclosure system that allows multi-ecosystem thermal data collection for environmental monitoring
- System currently used by Dr. Roydon Fraser for grad research at the University of Waterloo Environmental Reserve

RC Load Carrying Robot

SolidWorks, Machine Tools, Arduino, ESP32

- Designed and built a remote control robot capable of carrying over five times its weight

Education

University of Waterloo

BASc in Mechanical Engineering
Sep. 2021 – June 2026

Experience

Toyota Motor Manufacturing Canada

Sep – Dec 2025

Engineering Analyst Intern - Stamping

Cambridge, ON

- Led mechanical design for die arm testing system using **Solidworks** and applied **FEA** for design validation to achieve in-house die arm troubleshooting
- Performed **Root Cause Analysis** on stamping line, creating **SOPs** to decrease line downtime by 16%
- Created 2D engineering drawings using **GD&T** to control position variability
- Managed **Steel Cost Reduction** project to reduce cost by over \$1,000,000 a year
- Created dashboards using **PowerApps** to identify and organize root causes of weekly downtime for engineering managers

Toyota Motor Manufacturing Canada

Jan – Apr 2024

Engineering Analyst Intern - Pilot Assembly

Cambridge / Woodstock, ON

- Designed assembly jigs and fixtures using **Solidworks** to be SLS 3D printed following **DFMA** guidelines
- Managed an **EVT** to optimize the hub nut tightener station that reduces total line downtime by 7%, producing a 96.1% yield output
- Conducted **feasibility study** on taillight to ensure correct assembly metrics before production
- Managed pull aheads such as AGV cage demolition and assembly line extension to support assembly transition

Mercuria

May – Aug 2023

Mechanical Commodity Specialist Intern

Beijing, China

- Investigated root causes for crude oil impurity by applying **Six Sigma DMAIC** principles to analyze supplier quality data
- Conducted supplier evaluations across Asian markets for mechanical equipment and process to support sourcing strategy for a multi-million dollar commodity portfolio

SWAP Robotics

Sep – Dec 2022

Operation Technician Co-op

Kitchener, ON

- Designed and built a charging station using **Solidworks** to charge 8 body cameras simultaneously, eliminating the need of human intervention
- Led robot assembly and oversaw tasks such as **wiring**, **soldering**, **PCB programming**, and overall robot assembly
- Created detailed **SOPs** to speed up robot functionality testing time by at least 50%

Multimatic Inc.

Jan – Apr 2022

Test Engineer Student

East Gwillimbury, ON

- Designed and constructed actuator setups using **Solidworks** to test control arm reliability
- Programmed and ran **reliability tests** such as UTS tests (including cyclic load cases), ball joint/bushing push outs, ball joint function fatigue tests, and torque measurements
- Analyzed and processed data collected from **DAQ** systems to look for failures and outliers
- Tested and supported all stages of **product development**, including **EVT**, **DVT**, **RVT**, **Ramp** and **MP** of upper control arm for a Chrysler vehicle